

Total No. of Questions : 6]

SEAT No. :

P5106

[Total No. of Pages : 4

BE/Insem.-506

B.E. (Civil Engineering)

**(c) ADVANCED CONCRETE TECHNOLOGY
(2012 Pattern) (Elective - I) (Semester - I)**

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) Answer Q.1 or 2, 3 or 4 and 5 or 6.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Your answers will be valued as a whole.
- 5) Use of electronic pocket calculator is allowed.
- 6) Assume suitable data, if necessary.
- 7) Use of IS code 10262,456 is not allowed.

- Q1)** a) Write a short note on manufactured sand as a fine aggregate. [4]
b) What do you mean by quality assurance and quality control? Give the IS recommendations of quality assurance. [6]

OR

- Q2)** a) What is heat of hydration? Explain the factors affecting heat of hydration. [4]
b) Explain workability as a quality measure of green concrete . On which factors workability of concrete depends? [6]

- Q3)** a) What is light weight concrete? How it can be achieved in practice? [4]
b) Write a short notes on : [6]
i) High strength concrete
ii) High performance concrete

OR

- Q4)** a) What are the different types of industrial waste materials useful for construction industry? Explain any one waste material based concrete in detail. [4]

P.T.O.

b) Write a short notes on :

[6]

- i) Jet cement concrete (Ultra rapid hardening concrete)
- ii) Vacuum concrete.

Q5) Using Indian Standard recommended guidelines, design a concrete mix for a reinforced concrete structure to be subjected to the very severe exposure conditions for the following requirements : **[10]**

A) Stipulations for proportioning

- a) Grade designation : M35,
- b) Standard deviation, $s = 5$
- c) Type of cement : OPC 53 grade conforming to IS 8112
- d) Maximum water-cement ratio : 0.45
- e) Workability : 75 mm(slump)
- f) Degree of supervision : Good
- g) Type of aggregate : Crushed Angular aggregate,
- h) Maximum cement content : 450 kg/m^3
- i) Minimum Cement content : 340 kg/m^3
- j) Method of concrete placing Pumping
- k) Chemical admixture type : Super plasticizer

B) Test data for materials

- a) Specific gravity of cement : 3.15
- b) Specific gravity of admixture : 1.145
- c) Specific gravity of
 - i) Coarse aggregate - 2.74
 - ii) Fine aggregate - 2.74
- d) Water absorption
 - i) Coarse aggregates - 0.5%
 - ii) Fine aggregates - 1.00%
- e) Free surface moisture
 - i) Coarse aggregates - Nil(absorbed moisture also nil)
 - ii) Fine aggregates - Nil

f) Sieve analysis

i) Coarse aggregate :

IS Sieve sizes (mm)	Analysis of Coarse Aggregate Fraction		Percentage of different Fractions			Remark
	I	II	I (60%)	II (40%)	Combined (100%)	Confirming of Table 2 of IS 383
20	100	100	60	40	100	
10	0	71.2	0	28.5	28.5	
4.75		9.40		3.7	3.7	
2.36		0				

ii) Fine aggregate : Conforming to grading zone I

C) Design considerations :

Table 1 : From IS 10262; Maximum water content per cubic meter of concrete

Sr. No.	Nominal Maximum Size of Aggregate (mm)	Maximum Water Content (kg)
i)	10	208
ii)	20	186
iii)	40	165

Table 2 : From IS 10262; Volume of Coarse Aggregate per Unit Volume of Total Aggregate

SI. No.	Nominal Maximum Size of Aggregate (mm)	Volume of Coarse Aggregate Per Unit Volume of Total Aggregate for Different Zones of Fine Aggregate			
(1)	(2)	Zone IV	Zone III	Zone II	Zone I
i)	10	0.50	0.48	0.46	0.44
ii)	20	0.66	0.64	0.62	0.60
iii)	40	0.75	0.73	0.71	0.69

OR

- Q6)** a) Write a short note on ground penetration radar technique. [4]
- b) Write a short notes on non destructive testing methods. [6]
- i) Stress wave propagation method
 - ii) Nuclear method.

